

Daniel Park

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EDUCATION

Western University

*Bachelor of Science, specialization in **Computer Science***

- **Relevant Coursework:** Machine Learning, Databases, Operating Systems, Computer Networks, Software Engineering

Certifications: AWS Machine Learning Engineer Certificate, Meta Back-End Developer Specialization Certificate

EXPERIENCES

Machine Learning Intern

Sept. 2025 – Apr. 2026

HEAL (Python, AWS S3, Airflow, Google Places API, Scikit-Learn, XGBoost)

- Architected a **cloud-native ETL pipeline** on **AWS S3** via **Apache Airflow** and the **Google Places API** to ingest logs into a Data Lake and detect **spatial anomalies**, processing **25K+** retail records across **80+** Canadian cities.
- Engineered **Python** workflows to sanitize **unstructured text** and normalize **metadata**, reducing **dataset anomalies** and inconsistencies by **30%**.
- Implemented **MLOps practices**, **versioned storage**, and rigorous **schema validation** to establish **data governance** and ensure pipeline reliability.
- Trained **predictive Scikit-Learn** and **XGBoost** models (**clustering, time-series forecasting**) to analyze **retail density trends**, driving end-to-end Model Deployment and analytics dashboards for **data-driven** strategic decisions.

Data Science Intern

Sept. 2024 – Aug. 2025

interRAI (Python, PySpark, Spark SQL)

- Managed **distributed PySpark ETL** workflows to validate and transform **250K+** clinical data assessment records across healthcare informatics infrastructure, ensuring **high-fidelity inputs** for health research.
- Modernized legacy **data pipelines** into **automated Spark SQL workloads**, reducing execution runtime by **30%** and lowering manual processing overhead.
- Documented core **Spark architectures** and led technical training, accelerating onboarding and streamlining **cross-functional collaboration**.

Software Developer Intern

May 2024 – Aug. 2024

BERL (React, FastAPI, CSS Flexbox/Grid, Cascade CMS)

- Refactored **15+** legacy **React components** via **CSS Flexbox/Grid** to cut page latency while building **6+ FastAPI endpoints** for administrative workflows enforcing accessibility standards.

Software Developer Intern

Sept. 2023 – Apr. 2024

Cultureplex (Python, SQL, PostgreSQL, LLMs, Matplotlib, Seaborn)

- Engineered **Python** and **SQL ETL pipelines**, leveraging **LLMs (Generative AI)** to transform **1K+** unstructured research records into **searchable metadata** and **structured PostgreSQL tables** for Data Warehousing.
- Optimized **PostgreSQL databases** via advanced **data modeling, indexing, and query tuning**, improving **retrieval latency** by **35%**.
- Generated **Matplotlib** and **Seaborn data visualizations**, alongside Tableau and PowerBI reports, to translate **aggregated data** for **cross-functional teams**, accelerating academic publications.

PROJECT

Canadian Ski Resorts Powder Predictor | *Airflow, PySpark, AWS S3/RDS, XGBoost, Flask, React*

- Architected an **Airflow, AWS S3, and PySpark ETL pipeline** utilizing a Data Lakehouse with **CI/CD** to ingest **62K+** records and automated **7-day forecasts** via **Open-Meteo REST APIs** for **18** Canadian ski resorts.
- Engineered temporal features, including **3-day rolling snowfall** and freeze-thaw cycles, to train an **XGBoost Regressor** predicting a "Powder Score" (**1-10**) with an MAE of **0.01**, storing inferences in an **AWS RDS PostgreSQL** warehouse.
- Built a **Flask REST API** and a dark-mode **React** dashboard to expose powder scores, resort comparisons, and weather-driven recommendations across **15+** resort views.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C/C++, C#, JavaScript, Bash, PHP, HTML/CSS

ML & Data: PySpark, Spark SQL, Airflow, Data Warehousing, Data Lake / Lakehouse, CI/CD, Model Deployment, pandas, NumPy, Polars, SciPy, Scikit-Learn, XGBoost, PyTorch, TensorFlow, LLMs / Generative AI, Matplotlib, Seaborn, SHAP, UMAP, Yellowbrick, Jupyter, ETL/ELT, Tableau, PowerBI

Backend & Cloud: AWS S3/RDS, Azure, GCP, Flask, FastAPI, REST/RESTful APIs, Django, .NET, React, PostgreSQL, MySQL, MSSQL, SQLite, MongoDB, DynamoDB, Snowflake, Elasticsearch, Lucene, MapReduce, Docker, Git, Maven, Regex